EECE 430 — Software Engineering

Final Examination

Name: ______________________________ Date: ____________________

Student ID: __________________________ Section: __________________

Instructions:

- This exam contains **33 multiple-choice questions**.
- Time allowed: **1 hour**.
- Circle the letter of the **best** answer for each question.
- Each question is worth equal marks.
- No electronic devices, notes, or textbooks are permitted.

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1. Which of the following best describes why agile methods propose constant code refactoring?

A) To ensure all documentation is kept up to date
B) To make future changes easier since they cannot be reliably anticipated
C) To eliminate the need for software testing
D) To increase the number of features delivered per sprint

2. In Test-Driven Development (TDD), what happens immediately AFTER a developer writes a new test?

A) The developer refactors existing code
B) The developer runs the test, which is expected to fail because the functionality is not yet implemented
C) The developer writes documentation for the new functionality
D) The developer merges the code into the main branch

3. A team uses the Fibonacci scale for story point estimation. Two team members estimate a user story as 2 and 13 respectively. According to agile estimation practices, what should happen next?

A) Average the two estimates and assign 7.5 story points
B) Use the higher estimate to be safe
C) Discuss the disagreement further, as the large gap suggests different understandings of the story
D) Assign the story to the person who estimated 2 since they are likely more skilled

4. Which statement about pair programming is INCORRECT?

A) Pairs are created dynamically so all team members work with each other
B) It develops common ownership of code and spreads knowledge across the team
C) It is always less efficient than two programmers working separately
D) It serves as an informal review process since each line of code is seen by more than one person

5. Which of the following is a limitation of Test-Driven Development?

A) It does not produce any form of system documentation
B) It eliminates the need for regression testing
C) Some tests can be very difficult to write incrementally, and it is hard to judge test completeness
D) It prevents developers from writing automated tests

6. In Scrum, what is the PRIMARY purpose of the daily scrum (stand-up) meeting?

A) To assign tasks to individual developers for the day
B) To share information, review progress, and identify problems so the team can re-plan short-term work
C) To demonstrate completed features to the Product Owner
D) To refine and estimate product backlog items

7. A Scrum team has a velocity of 21 story points per sprint. During sprint planning, the Product Owner wants to include PBIs totaling 35 story points. What should happen?

A) The team should extend the sprint length to accommodate all 35 points

B) The team should commit to 35 points and work overtime if needed

C) The team should select approximately 21 points of PBIs based on priority, and return the rest to the backlog

D) The ScrumMaster should override the velocity estimate and accept all items

8. Which of the following is TRUE about Extreme Programming (XP)?

A) XP mandates that increments are delivered to customers every 6 months

B) XP requires that all tests must pass for every build, and the build is only accepted if tests run successfully

C) XP discourages pair programming in favor of individual code ownership

D) XP was designed to integrate easily with traditional management practices in most organizations

9. A Product Backlog Item (PBI) that is described as a high-level idea that may radically change or may not be included in the final product is in which state?

A) Ready for implementation

B) Ready for refinement

C) Ready for consideration

D) Ready for deployment

10. Which of the following is NOT one of the top 5 benefits of Scrum mentioned in the course?

A) Unstable requirements do not hold up progress

B) The product is broken down into understandable chunks for stakeholders

C) It eliminates the need for a product backlog

D) Team communication is improved and everyone can see everything

11. An architect must optimize a system for BOTH maintainability and performance. Why is this inherently challenging?

A) Maintainability requires a centralized architecture while performance requires distribution

B) Maintainability favors fine-grained, replaceable components, but many components communicating slows the system down

C) Performance and maintainability both require the same architectural choices

D) Maintainability can only be achieved in layered architectures

12. In a multi-tier client-server architecture, which tier is responsible for implementing application-specific logic as well as information storage and retrieval requests?

A) The web server tier

B) The database server tier

C) The application server tier

D) The client tier

13. A team is designing a system where large volumes of information are generated by various software tools (e.g., code editors, debuggers, report generators) and must be shared among them. Which architectural pattern is MOST appropriate?

A) Pipe and Filter

B) Client-Server

C) Repository

D) MVC

14. Which statement about the pipe and filter architectural pattern is FALSE?

A) It is commonly used in batch data processing applications

B) Each filter carries out one type of data transformation

C) It is highly suitable for interactive systems

D) Evolution by adding transformations is straightforward

15. In a service-oriented architecture, services are described as 'stateless.' What does this mean and why is it important?

A) Services do not store any data at all, reducing storage costs

B) Services do not maintain internal state, allowing them to be replicated and migrated between servers

C) Services cannot access databases, ensuring complete independence

D) Services must restart from scratch after every request, improving security

16. A SaaS provider needs to allow different business customers to add custom fields to the shared database schema. Using the 'additional fields' approach, what is a key problem?

A) It requires each customer to have their own database instance

B) It is difficult to know how many extra columns to include, and different customers may need different data types

C) It eliminates the ability to use tenant identifiers

D) It prevents any form of database extensibility

17. Which of the following is an advantage of multi-tenant databases over multi-instance databases?

A) Complete data isolation between customers

B) Flexibility to customize database schemas per customer

C) Easier update management since only a single instance needs updating

D) Stronger resilience because a failure affects only one customer

18. A company is choosing between deploying their SaaS product on containers vs. virtual machines. Which scenario BEST justifies using containers?

A) The system requires complete hardware-level isolation between tenants

B) Users work independently with little data sharing, and services need to scale rapidly with minimal overhead

C) The system needs to run different operating systems for different services

D) The company prefers slower but more secure deployment

19. In cloud computing, what is the key difference between scalability and elasticity?

A) Scalability refers to handling more users; elasticity refers to handling fewer users

B) Scalability is the ability to cope with increasing load; elasticity also includes the ability to scale DOWN as demand decreases

C) They are identical concepts with different names

D) Elasticity only applies to private clouds while scalability applies to public clouds

20. Which of the following is NOT a SaaS design issue discussed in the course?

A) Local vs. remote processing trade-offs

B) Authentication and federated identity

C) Choosing between compiled and interpreted programming languages

D) Information leakage between organizations in multi-tenant systems

21. In continuous integration, what is the recommended 'integrate twice' approach to avoid breaking the build?

A) Push code to two separate repositories simultaneously

B) Integrate and test locally on your own machine first, then push to the shared repository

C) Have two different team members review the code before pushing

D) Run integration tests twice on the CI server for redundancy

22. Which of the following correctly describes the relationship between continuous delivery and continuous deployment?

A) They are the same thing with different names

B) Continuous delivery tests the system in a production-replica environment; continuous deployment automatically releases it to users

C) Continuous deployment tests the system; continuous delivery releases it

D) Continuous delivery only applies to containerized applications

23. A build system checks the modification timestamps of source files and compiled files. If the source file's timestamp is NEWER than the compiled file's timestamp, what does the build system do?

A) It does nothing because the compiled code is up to date
B) It recompiles the source file to create an updated compiled version
C) It deletes the compiled file and alerts the developer
D) It reverts the source file to match the compiled version

24. Which characteristic of Infrastructure as Code ensures that infrastructure changes can be rolled back if problems arise?

A) Visibility
B) Reproducibility
C) Recovery — infrastructure models can be versioned and stored in a code management system
D) Reliability

25. According to the IBM DevOps metrics scorecard, which combination of metric trends indicates a healthy DevOps process?

A) Increasing failed deployments, decreasing deployment frequency
B) Decreasing failed deployments, decreasing mean time to recovery, increasing deployment frequency
C) Stable deployment frequency, increasing lead time from development to deployment
D) Decreasing customer numbers, increasing availability

26. A microservice is described as having 'high cohesion and low coupling.' Which of the following violates BOTH of these principles?

A) A payment service that handles only payment processing with its own database
B) A service that handles user registration, email notifications, and report generation, and directly accesses three other services' databases
C) A service that communicates with other services only through a message broker
D) A service that manages password creation, validation, recovery, and storage

27. A microservices system uses eventual consistency. Which statement BEST describes how this works?

A) All database replicas are always perfectly synchronized at all times
B) Database changes are recorded in a pending updates log; other service instances read the log, update their databases, and confirm the update
C) A central coordinator locks all databases during updates to prevent inconsistency
D) Each service ignores data from other services to maintain independence

28. What problem do circuit breakers solve in a microservices architecture that simple timeouts do not?

A) Circuit breakers encrypt service communications to prevent attacks
B) Circuit breakers immediately deny access to a failed service, avoiding the delay that timeouts cause on every call to a failed service
C) Circuit breakers eliminate the need for error handling in individual services
D) Circuit breakers allow services to share databases more efficiently

29. In the context of microservice coordination, what is the key disadvantage of choreography compared to orchestration?

A) Choreography requires a central controller that becomes a bottleneck
B) Choreography is harder to debug because there is no central point that tracks the workflow state
C) Choreography does not support asynchronous communication
D) Choreography requires all services to use the same programming language

30. An attacker uses a cross-site scripting (XSS) attack to steal a session cookie from a victim's browser. The attacker then uses this cookie to access the victim's account. This SECOND attack is an example of:

A) SQL injection
B) Brute force attack

C) Session hijacking
D) Denial of service

31. A system implements a policy that locks user accounts after 5 failed login attempts. Why can this security measure itself become a vulnerability?

A) Because it makes brute force attacks easier
B) Because attackers can intentionally trigger lockouts to deny legitimate users access (user lockout attack)
C) Because it eliminates the need for strong passwords
D) Because it conflicts with multi-factor authentication

32. In the context of data encryption, which statement about 'data at rest' vs. 'data in use' is CORRECT?

A) Data at rest should remain unencrypted for faster access; data in use should always be encrypted
B) Files storing data at rest should be encrypted; encrypting data in use slows response time and makes general search impossible
C) Both data at rest and data in use should never be encrypted to maintain performance
D) Data at rest encryption uses asymmetric keys while data in use encryption uses symmetric keys exclusively

33. According to data protection principles covered in the course, which of the following is a VIOLATION of proper data handling?

A) Allowing users to discover what personal data is stored about them
B) Storing user data in a country with weaker data protection laws without an agreement to uphold stronger rules
C) Deleting personal data when a user deletes their account
D) Obtaining user consent before disclosing their data to other people

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— END OF EXAM —